

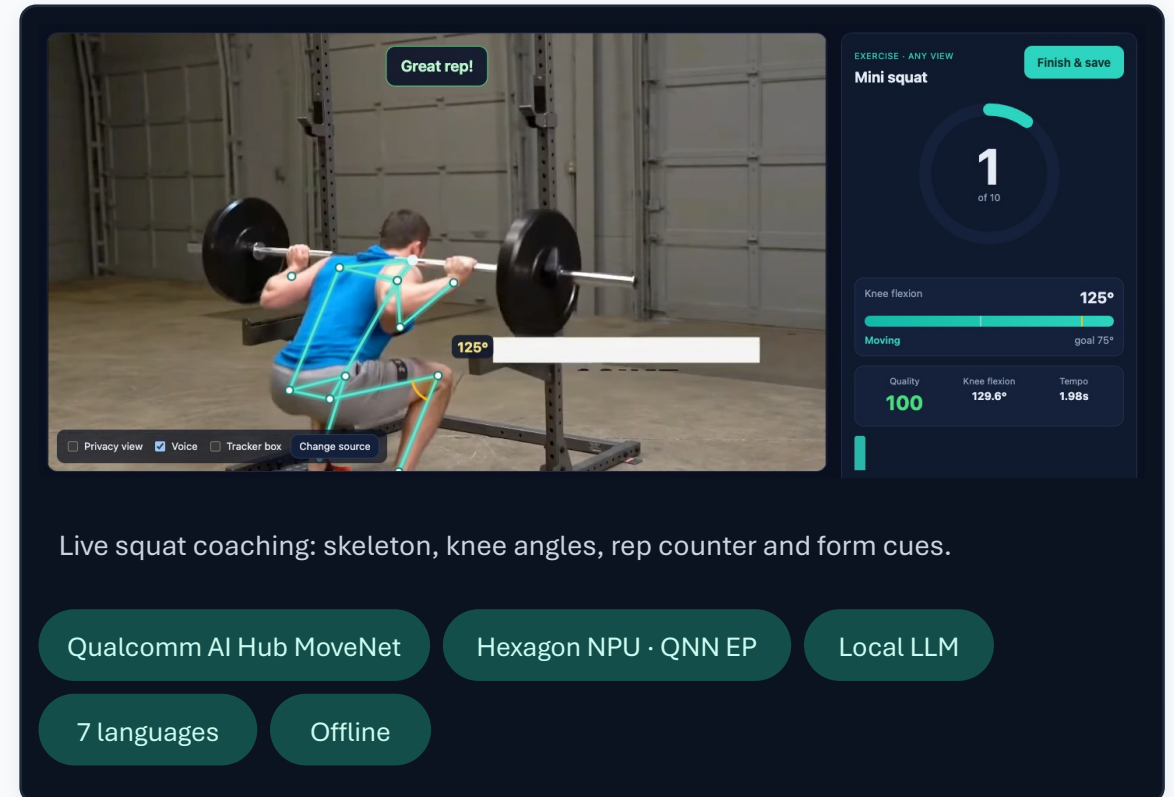
PunarGati

पुनर्गति · movement, restored

**A physiotherapist's eyes at home,
running on the Snapdragon NPU.**

On-device pose tracking, clinical screening tests and coaching in 7 Indian languages. Offline, private, and with no subscription.

Sachin · github.com/Sachin0496/punargati



THE PROBLEM

Recovery happens at home: unsupervised, uncounted, unmeasured

28.7%

pooled prevalence of knee osteoarthritis in Indian studies

Pal et al., Indian J Orthop 2016

101 M

Indians with diabetes. About 13% of them develop frozen shoulder

ICMR-INDIAB 2023 · Zreik et al. 2016

1 in 3

adults over 65 fall every year, and fall-risk screening needs a trained assessor

WHO Falls Prevention Report 2007

Patient gets an exercise sheet and a weekly visit

Nobody counts reps, checks form or measures range at home

The physio finds out a week later, or never

A camera in someone's bedroom must never stream to a server

	Cloud AI coach	PunarGati on a Snapdragon PC
Privacy	Home video of patients goes to a server: a DPDP-Act risk	Frames live in RAM for ~1 ms on the NPU. Only numbers are saved; there's a skeleton-only privacy view
Connectivity	30 fps upload fails in tier-2/3 towns	Fully offline. Models ship inside the repo
Cost	Per-minute inference makes free home rehab impossible	One-time laptop, unlimited sessions
Latency	A network round trip breaks rep timing and live cues	~1 ms pose inference and instant spoken feedback

PunarGati turns a Snapdragon laptop into a physiotherapist's eyes

1

Coach

9 prescribed exercises. Reps per side, range of motion, tempo and a quality score for every rep. Spoken form corrections.

2

Test

30-s chair stand (CDC fall-risk norms), single-leg balance, and shoulder / knee range of motion against AAOS reference values.

3

Track

Recovery curves per joint, day streaks, and a printable report the patient hands to the physio.

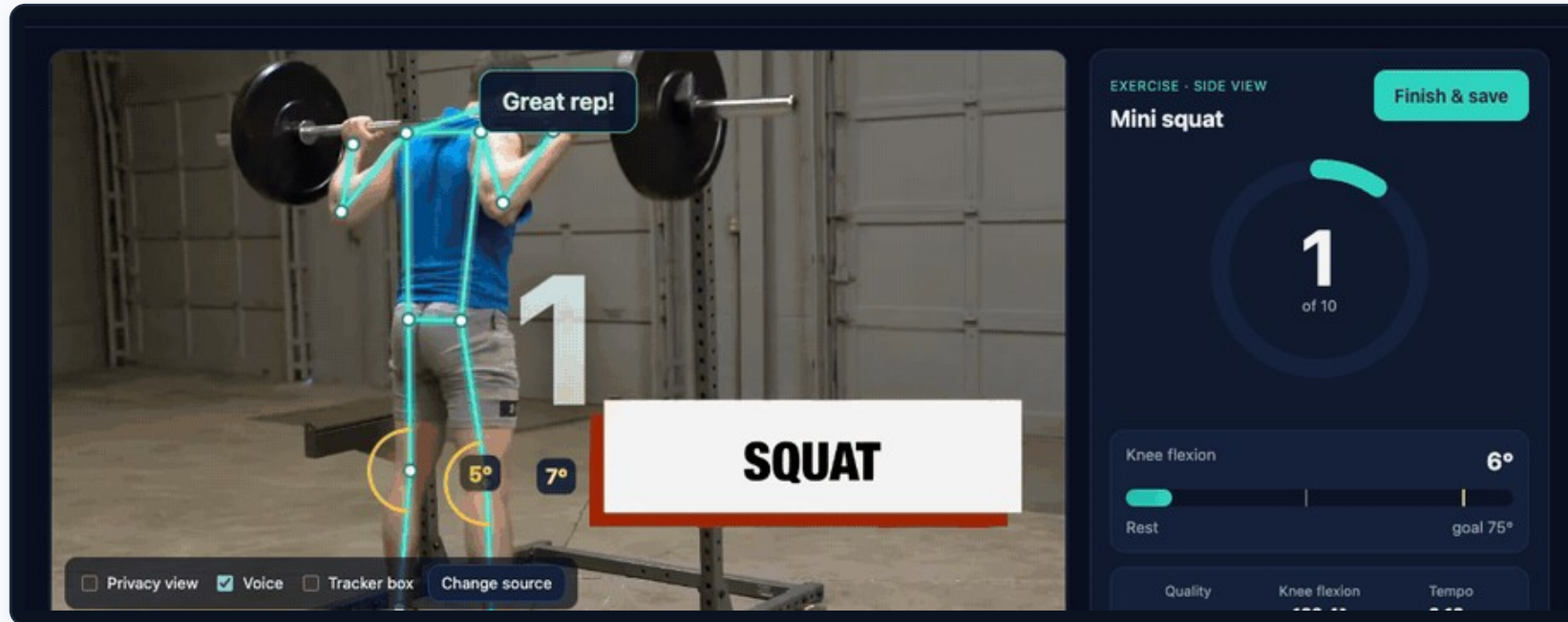
4

Plan & ask

Paste the physio's note to get a one-tap plan. The local LLM answers questions from your own data and escalates red flags.

Coaching languages: English, Hindi, Tamil, Telugu, Kannada, Marathi and Bengali, with offline Windows voices

Every rep measured: angle, tempo and quality, plus a spoken correction



Goniometer-style joint angles

Knee, hip, shoulder, elbow and trunk, computed in isotropic pixel space

Per-rep quality score

Range reached × tempo × form, so a partial rep gets "go a little further"

View-aware form rules

Knees caving in (front view), chest dropping (side view), elbow bending, shrugging

Privacy view

Hide the room and show only the skeleton

Validated on real footage: 3/3 stands on the CDC's own chair-stand video (front view) and 2/2 squats on the demo clip. Both are regression tests.

Standard fall-risk and mobility tests, scored against published norms

30-second chair stand

Counts full stands in 30 s and flags fall risk using CDC STEADI thresholds by age and sex

Single-leg balance

Times the stance until the foot touches down. Under 5 s flags fall risk (Vellas 1997); age means from Springer 2007

Range of motion

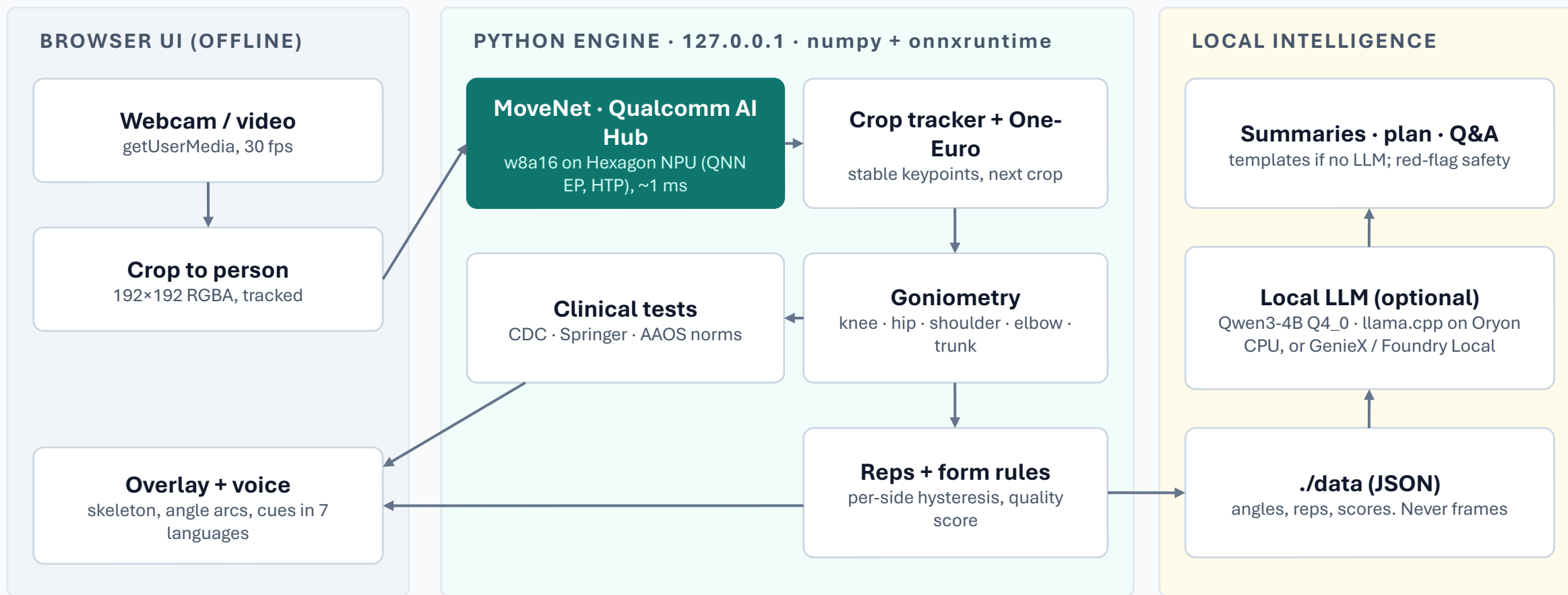
Shoulder flexion and abduction, knee flexion. Median-of-5 frames, compared with AAOS 180° / 180° / 135°

30-s chair stand: stands that indicate fall risk (CDC STEADI)

Age	Men	Women
60–64	< 14	< 12
65–69	< 12	< 11
70–74	< 12	< 10
75–79	< 11	< 10
80–84	< 10	< 9
85–89	< 8	< 8
90–94	< 7	< 4

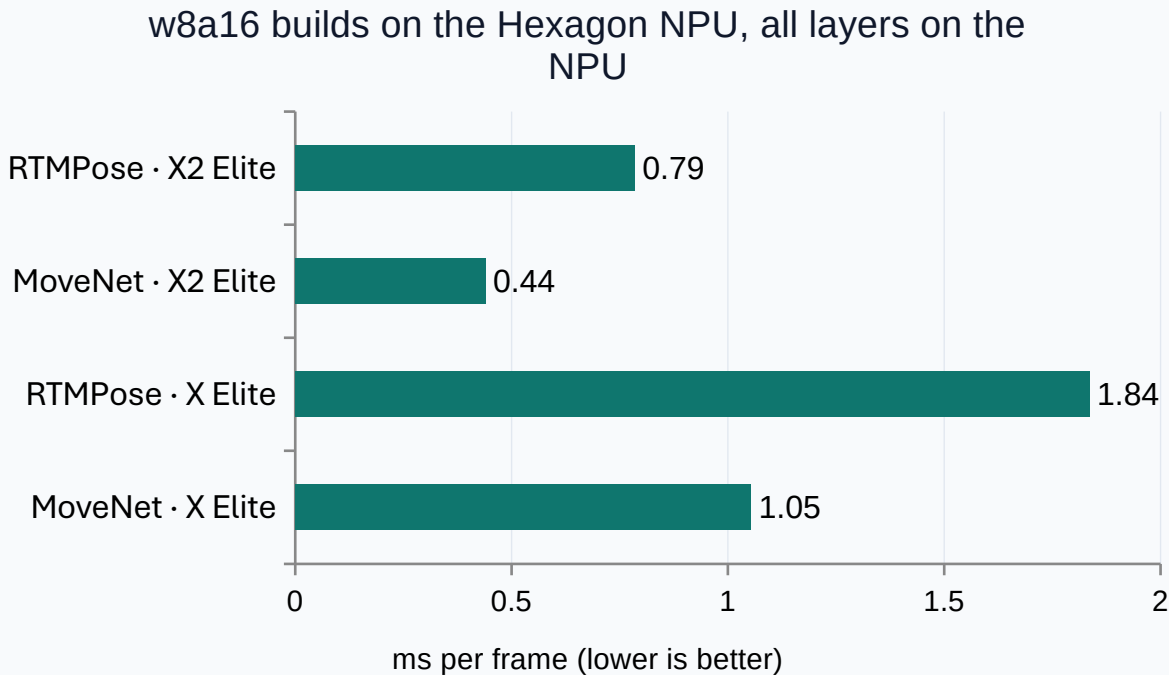
Automatic countdown, timing and counting. The result card compares the score with the patient's age band.

The browser sees, the NPU understands, and nothing leaves the PC



Only 3 native dependencies (numpy, onnxruntime, onnxruntime-qnn), all with Windows-on-ARM64 wheels. No OpenCV, PyTorch or npm build.

Two AI Hub pose models run on the Hexagon NPU in under 3 ms per frame



Source: Qualcomm AI Hub device profiles, v0.63.0 (ONNX). The cascade uses <10% of the 33 ms frame budget at 30 fps.

Pose cascade

Hexagon NPU

MoveNet tracks and RTMPose-WholeBody refines (133 keypoints, feet). onnxruntime-qnn, strict 100% offload, context cache

Benchmark / fallback

Adreno GPU · CPU

Same model through the QNN GPU backend or the ORT CPU EP, with the fallback reason shown in the UI

LLM, after a session

Oryon CPU

llama.cpp Q4_0 with i8mm repack. It decoded faster than the NPU/GPU backends in our earlier X Elite tests

Built-in benchmark (NPU vs GPU vs CPU): latency, CPU load at 30 fps, and battery watts on the user's own device.

Engineered to be accurate, provable, and impossible to break in a demo

Provable NPU offload

Sessions are built with `session.disable_cpu_ep_fallback=1`. Success means every operator is on the Hexagon NPU.

Two-model NPU cascade

MoveNet tracks the person. RTMPose-WholeBody re-estimates 133 keypoints on its box. Exact uint16 I/O from AI Hub metadata.

Accuracy from small models

MoveNet crop tracking keeps the person large; One-Euro filtering in body-scale units.

Honest rep counting

Per-side hysteresis state machines, duration/timeout guards, form rules that must persist 4 frames.

View-independent tests

Self-calibrating shoulder-height signal for sit/stand. 3/3 stands on the CDC's own front-view video.

Never dead in a demo

NPU → GPU → CPU ladder, a QNN 1.x fallback route, LLM-free fallbacks, 31 tests incl. real video.

One script to install, one click to run, and no internet after that

1

scripts\setup.ps1

Finds or installs native ARM64 Python, installs 3 packages, verifies AI Hub models by SHA-256, and self-tests the NPU

2

PunarGati.bat

Double-click. The browser opens on 127.0.0.1 and the NPU badge shows the live latency

3

scripts\setup-llm.ps1 (optional)

llama.cpp win-arm64 + Qwen3-4B Q4_0. Auto-detected, or use GenieX, Foundry Local or LM Studio

Accessible by default

- 7 coaching languages + Hindi interface
- Privacy view: skeleton only
- Runs on any laptop (CPU fallback)
- Bundled demo clip, no webcam needed
- No account, no subscription
- Data stays in ./data on the PC

Tested on every push by CI on a native Windows-on-ARM64 runner: install script, QNN plugin + fallbacks, local-LLM scripts and all tests.

Next: hands, feet and voice, all on the same NPU



Patients

supervised-quality home exercise, visible progress, early fall-risk flags

Physiotherapists

measured adherence and ROM trends between visits

Tier-2/3 India

works with no connectivity and no cloud bill

Private. Offline. Multilingual. Measured.

PunarGati puts a physiotherapist's eyes in every home that has a Snapdragon PC.

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